Nikhil Singh

Backend & Platform Engineer. Rust, Go, Nix, distributed systems, payment infrastructure, developer tooling.
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Experience

Product Engineer | *Juspay Technologies* Jun 2025 – Present

- Architected and solely own **Project Bernoulli**, a Rust UPI payment gateway bridging PSP SDK traffic (Newton to BHIM), handling 2M+ daily transactions with 99.99% uptime.
- Built UPI Mandate interoperability, Tap & Pay, and Unified Agentic Payments (UAP); launched Contextual Payments with BHIM at Global Fintech Fest across 4 merchants, incl. Zepto and Vi.
- Helped adopt active-active cell architecture across AWS and GCP with per-cloud Kubernetes clusters, enabling zero-downtime failover with sub-3-minute replication lag.
- Migrated **ProcessTracker**, a Haskell scheduler with custom retries and interoperable Kafka/Redis queues, from Stack to Nix flakes with reproducible builds and cached CI.
- Engineered layered Jenkins CI, cutting bootstrap from 20–40 min to 2–5 min and builds by 40–60%; deployed a Nix devbox adopted by 800+ engineers.

Linux Systems Intern | *Decet Technology* 2023 – 2024

- Slashed deployment times from 2+ hours to under 5 minutes by engineering a Go-based CI/CD system.
- Built Python backend services handling 10K+ daily requests with REST APIs and database integrations.

Nixpkgs Maintainer & Open Source Contributor | *NixOS & nix-community* 2023 – Present

- Registered nixpkgs maintainer of 3 packages, incl. minimal-tmux-status (275-star tmux theme) and fzf-preview; 60+ merged PRs across 30+ open-source projects (nixpkgs, home-manager, nixvim, hyprwm/contrib).
- Upstreamed Nix builds and modules to Juspay OSS: migrated Hyperswitch from cargo2nix to rust-flake, contributed to omnix and services-flake, and added home-manager modules (pi-coding-agent, sesh).

Projects

Ndots | *NixOS System Configuration (119+ Stars)* 2024 – Present

- Modular multi-host NixOS/nix-darwin flake with centralized user/SSH management and reusable, auto-wired modules.
- Automated remote provisioning with nixos-anywhere and CI-built x86_64/aarch64 installer ISOs.

Nvix | *Nixvim/Neovim Distribution (174+ Stars, 42+ Forks)* 2024 – Present

- Modular Nixvim distribution offering bare, core, and full variants for different workflows.
- Auto-discovered plugin modules can be cherry-picked, extended, or consumed independently.

Skills & Open Source

Languages: Rust, Go, Python, Haskell, SQL, Bash
Infrastructure: Kubernetes, Docker, AWS, GCP, Kafka, ClickHouse, PostgreSQL, Redis, CI/CD
Nix/Linux: Nix/NixOS, nixpkgs, flakes, Home Manager, overlays, modules, sops-nix, Linux
Open Source: Member of NixOS, nix-community, nixos-asia

Education & Publications

Gautam Buddha University Greater Noida, UP
Master of Technology in Software Engineering ***(CGPA: 8.83)*** *Aug 2023 – May 2025*
Bachelor of Technology in Computer Science Engineering *Aug 2020 – Aug 2023*

Publications:
"Modularity and Lazy Evaluation in NixOS: A Functional Approach to Operating System Configuration," IEEE.
"ML-based Heavy Rainfall Prediction for the Northern Himalayan Region," IEEE (co-authored).